

BLUESTOCK FINTECH CAPSTONE PROJECT

Mutual Fund Analytics Platform

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Data Analyst Internship — Final Project Report

1. Executive Summary

This project aims to build a full-stack Mutual Fund Analytics Platform for Bluestock Fintech. Using Python (Pandas), an ETL pipeline was developed to ingest, clean, and transform public AMFI data. The processed data was loaded into an SQLite database with a star schema and connected to an interactive Power BI dashboard. The platform provides deep insights into AUM trends, investor demographics, and advanced risk metrics, helping users make data-driven investment decisions.

2. Data Sources & Datasets

The project utilized 10 CSV datasets provided by Bluestock Fintech, sourced from publicly available AMFI India data and mfapi.in. These datasets cover 40 mutual fund schemes, historical daily NAV for 4.5 years, quarterly AUM data, monthly SIP inflows, investor transaction logs (32,000+ rows), and benchmark indices.

3. ETL Pipeline & Database Design

An ETL pipeline (etl_pipeline.py) was built to handle data ingestion, cleaning, and validation. Data cleaning tasks included parsing dates, forward-filling missing NAV values (to handle weekends/holidays), and removing duplicates. The cleaned data was loaded into a normalized SQLite database (bluestock_mf.db) featuring a 5-table star schema (dim_fund, fact_nav, fact_transactions, etc.) to ensure efficient querying.

4. Exploratory Data Analysis (EDA)

Extensive EDA was performed in Jupyter notebooks to uncover trends. Key findings include a steady growth in SIP inflows up to ₹31,002 Crore in Dec 2025. AUM analysis revealed the dominance of SBI Mutual Fund (₹12.5L Cr), and demographic analysis showed that the 26-35 age group contributes the highest total transaction volume.

5. Performance & Risk Analytics

Financial metrics were computed for all funds, including Annualized CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown against the Nifty 100 benchmark. A composite fund scorecard was generated to rank the top-performing funds based on returns, risk-adjusted returns, and expense ratios.

6. Interactive Dashboard

An interactive Power BI dashboard (bluestock_mf.pbix) was designed with 4 functional pages covering Industry Overview, Fund Performance, Investor Analytics, and SIP/Market Trends. It includes KPI cards with Indian Rupee formatting and interactive slicers for Fund House, Age Group, and Date.

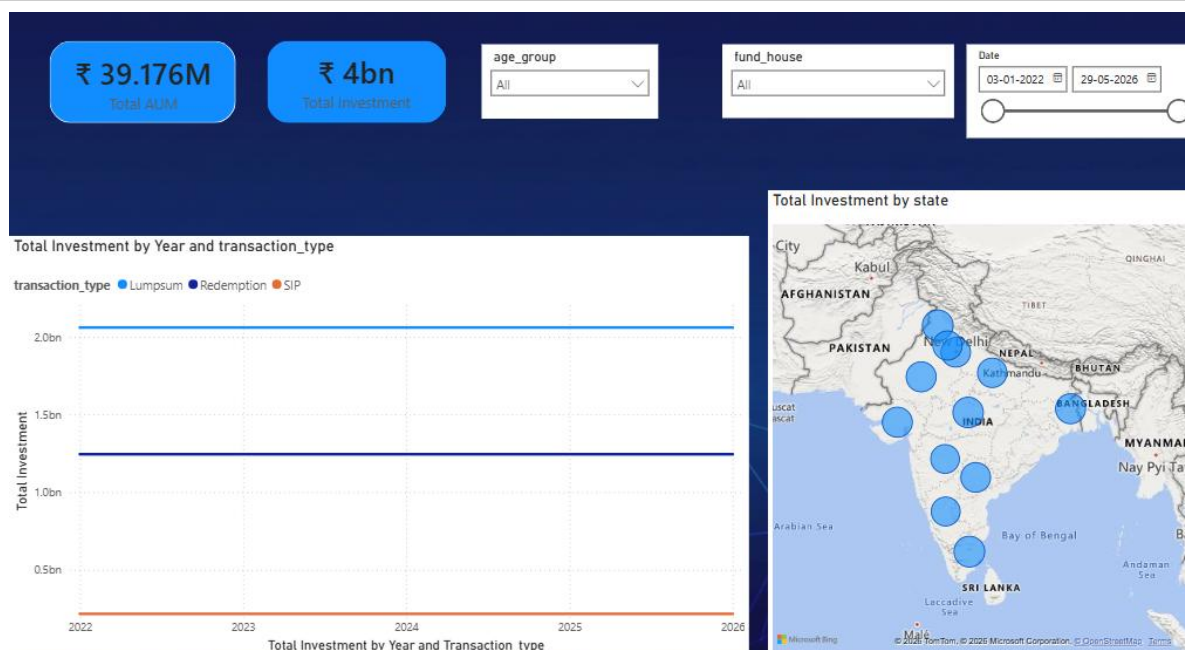


Figure 1: Industry Overview page — KPI cards, investment trend, and state-wise map.

7. Advanced Analytics

Advanced metrics were calculated, including Historical VaR (95%), rolling Sharpe ratios, and investor cohort analysis. A simple Python recommendation engine (recommender.py) was built to provide the top 3 funds based on the user's risk appetite (Low, Moderate, High).

8. Conclusion & Recommendations

The project successfully met all objectives by creating a unified data platform for mutual fund analysis. It is recommended to schedule the ETL pipeline to run daily to keep the data live, and to expand the investor transaction data to include more diverse demographics across India for deeper analysis.